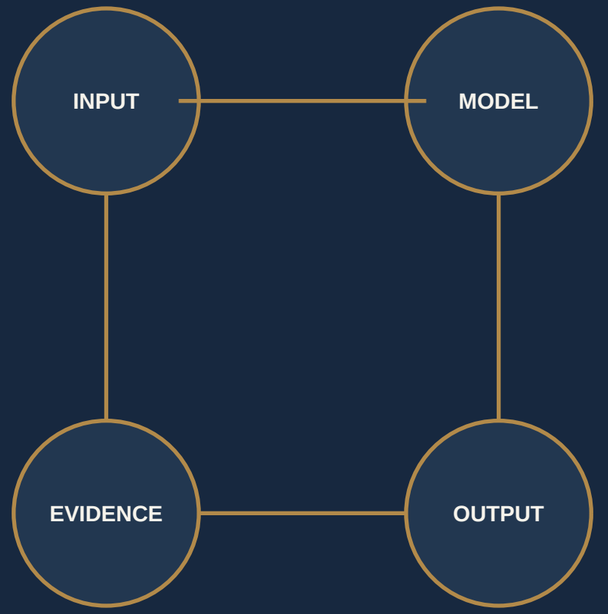
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Intelligent Control Systems

AI.308 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define ai-enhanced control using fuzzy logic.
- Apply key methods including neural networks.
- Build or analyze a practical system involving evolutionary algorithms.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

AI-enhanced control using fuzzy logic

02

METHODS

Neural networks

03

SYSTEMS

Evolutionary algorithms

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · AI-enhanced control using fuz

AI-enhanced control using fuzzy logic

02 · Neural networks

Neural networks

03 · Evolutionary algorithms

Evolutionary algorithms

04 · Reinforcement learning and ada

Reinforcement learning and adaptive systems

05 · Mandatory practical: Design an

Mandatory practical: Design and compare intelligent controllers on simulated robotic systems

Core ideas, made explicit

AI.308 • Distinguish definitions, assumptions and evidence

- AI-enhanced control using fuzzy logic
- Neural networks
- Evolutionary algorithms
- Reinforcement learning and adaptive systems
- Mandatory practical: Design and compare intelligent controllers on simulated robotic systems

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.308 • Distinguish definitions, assumptions and evidence

- AI-enhanced control using fuzzy logic
- Neural networks
- Evolutionary algorithms
- Reinforcement learning and adaptive systems
- Mandatory practical: Design and compare intelligent controllers on simulated robotic systems

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

AI-enhanced control using fuzzy logic

INPUTS

Neural networks

METHOD

Evolutionary algorithms

OUTPUT

Reinforcement learning and adaptive systems

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving ai-enhanced control using fuzzy logic.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using ai-enhanced control using fuzzy logic.
- Implement or simulate a baseline using neural networks.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.